<Add your name and student ID (SID) to the footer. Add your answers below. Include any relevant output from Excel with your answer. To receive full marks for probability questions, include your working as well as the final answer e.g. a copy of the formula used in Excel. Some questions may be easier to handwrite. In those cases scan in your answer and include your scan at the relevant point.>

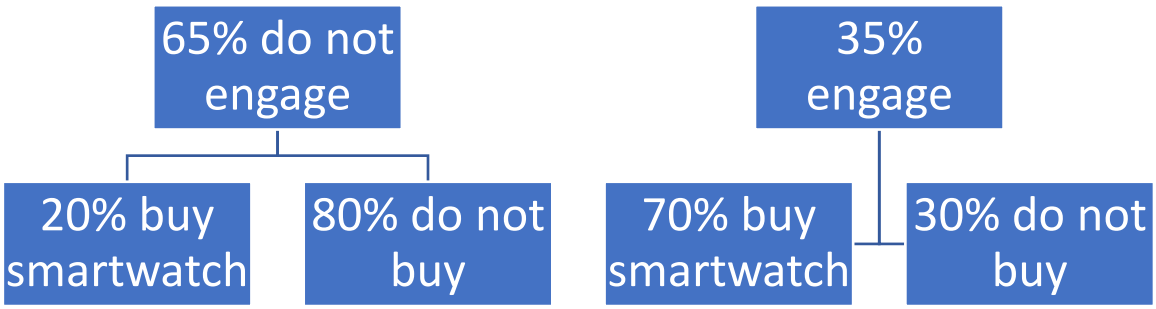
<Unless otherwise stated, proportions/probabilities should be written as a percentage to 1 decimal place (% 1dp for short). For instance, if a probability is 0.1234… then write this as 12.3%.>

Question 1: (16 marks)

a) A tech retailer is trialling a new promotional strategy for selling ~~sandwiches~~ smartwatches. Data shows that 35% of customers engage with the promotional display. Of those who engage, 70% purchase a smartwatch. In comparison, of those who do not engage, 20% still purchase a smartwatch,

i) Construct a fully labelled tree diagram to represent the above information. Let E = customer engages with the display, P= customer buys a smartwatch. **Hint**: check your final probabilities add to 1.

(3 marks)



ii) Identify the branches on your tree diagram where a customer will purchase a smartwatch. Hence determine the probability a customer buys a smartwatch.

(2 marks)

I need to sum the branches where people buy the smartwatch. It would be:

0.35*0.7+0.65*0.2 = 0.375= 37.5%

37.5% will buy a smartwatch

iii) Calculate the proportion of smartwatch purchases that were from customers who engaged in the display. Hint: this is equivalent to finding the probability that a customer engaged in the display given they made a purchase.

From 37.5% 0.35*0.7= 0.245 bought and engaged in the display.

Now:

0.245/0.375=0.6533

Approximately 65.33% of people that bought the smartwatch watched the display.

(2 marks)

iv) Can you use this data to suggest the promotion was successful or not? If yes, explain how it does. If not, explain why not. (2 marks)

In this exercise by itself, it can be appreciated that customers who interact with the display have higher buy rate (70% buy rate) than customers that do not (20% buy rate). This suggest that this display is associated to an increase of sales.

However, this information by itself it is not enough to say if the promotion was successful or not, as customers who chose to interact are already more interested in buying. To confidently say whether it is successful or not, a comparison might be necessary to be analysed.

b) To inform future promotions, the company surveyed a sample of 1500 adults. The table below shows the gender and whether they have purchased a smart watch.

	Purchased (P)	Not Purchased ($\bar{P}$)	Totals
Male (M)	300	500	800
Female (F)	400	300	700
Totals	700	800	1500

i) Calculate the probability that a randomly selected customer has not purchased a smartwatch. (2 marks)

To do that,
Just 800/1500 = 0.53333… = 53.3%

53,3% customers have not purchased a smartwatch

ii) Calculate the probability that a randomly selected customer is male and has not purchased a smartwatch.

(2 marks)

500/1500= 0.33333… = 33.3%
33,3% is for male who ha not purchased a smartwatch

iii) Are the events 'an individual is female' and 'an individual has purchased a smartwatch' independent? Explain using probabilities.

(3 marks)

Gender and purchasing are dependent because the probability of purchasing changes when we know the person's gender.

From 1500 people, 700 bought. So P(Bought)= 700/1500= 0.467
Means, if I don't know nothing about the gender, there is a 46.7% of probability that someone purchase.
From 700 women, 400 bought the smartwatch. So, P(Women|Bought)= 400/700 = 0.571.
Means, if I know it is a women, then the probability of buying raise to 57.1%

So, P (Women|bought) ≠ P(bought)

It is not same.

Question 2 (20 marks)

A seafood company produces frozen crumbed fish fillets sold both in individual and bulk-buy packs. A large random sample of fillets shows that the actual weight of each fillet is **normally distributed** with a mean of **75g** and a standard deviation of **4g**. There have been complaints in social media about the fillets being too small.

Australian law requires truth in labelling (including weights) but does allow for a small margin of deficiency. This question is based on some parts of the simple description of the legislation.

a) To start, assume you are the quality control officer for the company. You will use Excel to see if the company is likely to pass an inspection or not and recommend actions to management.

The first task is to consider the weight of individual fish fillets.

i) Write down the probability distribution of these weights. In particular, state the **type** of distribution and write down the value of all the **parameter**(s) of this distribution. (You need all of these to enter into Excel.) (2 marks)

Question 2		
1.		
type:	Normal distribution	
Parameters:		
	Mean:	75 g
	Standard deviation:	4 g

ii) Consider fillets sold individually with an **advertised net content of 70g**. In Australia, manufacturers need to comply with one of the two methods (the manufacturer chooses which one). The first method is the 'average system'. Under that system (a) the average content of a sample must not be lower than the advertised amount, and (b) no individual can have a "shortfall greater than 5%".

In order to check the second part of the rule, use Excel to **calculate the probability that a fillet has a shortfall of more than 5%.**

Hints: determine what the shortfall of 5% corresponds to in grams (g) and thus the weight of a deficient product. Calculate a probability in Excel using the values from part (i).

The advertised amount is 70 g. A shortfall of 5% corresponds to 3.5 g, so a fillet is deficient if its weight is below 66.5 g.

(3 marks)

2.		
	normal distribution	
	0,01679331	1,70%

This means approximately 1.7% of fillets exceed the allowed shortfall.

iii) If there is a risk of non-compliance, there are several options available. One choice is to change the advertised amount. To explore this choice, **determine the weight that 99.99% of fillets exceed.** (3 marks)

To find the weight that 99.99% of fillets exceed, we calculate the 0.01st percentile of the normal distribution:

3.		
60,1239341	=	60,1 g

(I used NORM.INV formula in excel to calculate the percentile. NORM.INV(0.0001,75,4))

Therefore, approximately 99.99% of fillets weigh more than **60.1 g**.

iv) Write a few sentences to your manager. **Explain if an inspection is likely to comply with legislation or not. Make at least 2 recommendations to reduce any non-compliance risk.**

Hints: this question is about communicating the statistics/numbers in a way a manager will understand and be convinced by. You should refer to (ii) explicitly.

(3 marks)

According to the findings in part (ii), the likelihood of a given fillet to have a shortage that exceeds 5 percent (i.e. less than 66.5 g) is around 1.7 percent. This implies that the majority of fillets will be able to meet the legislation, but a minor percentage will still provide an inspection failure.

To minimize the chance of non-compliance, the company might raise the target or advertised weight such that the number of fillets below the minimum acceptable limit is reduced. Also, better process control to minimize variation in production (smaller standard deviation) would further minimize the risk of underweight fillets. Such measures would enhance adherence, and minimize the chances of regulatory problems.

b) The manufacturer also produces bulk packs which package the fillets in **groups of 6, with an advertised net amount of 425 g**. The quality control officer has to determine if the advertised amount for the packs will comply with legislation. The second method to comply in Australia is called the Average Quantity System (AQS) for the packages. Rule 2 of AQS states that at most 6% of packages can be 'inadequate'. The AQS also defines what 'inadequate' is based on the advertised amount – in this case a bulk pack would be 'inadequate' if it is 12.8g or more below the advertised weight (12.8g is called the 'tolerance deficiency').

i) State the distribution, mean, and standard deviation of the <u>total</u> weight of a bulk pack (6 fillets).

Tips: Topic 8 gives the distribution for the <u>sample mean</u> of a group of size n. The <u>total</u> of the group has the same distribution, except the mean and standard deviation are multiplied by n.
(2 marks)

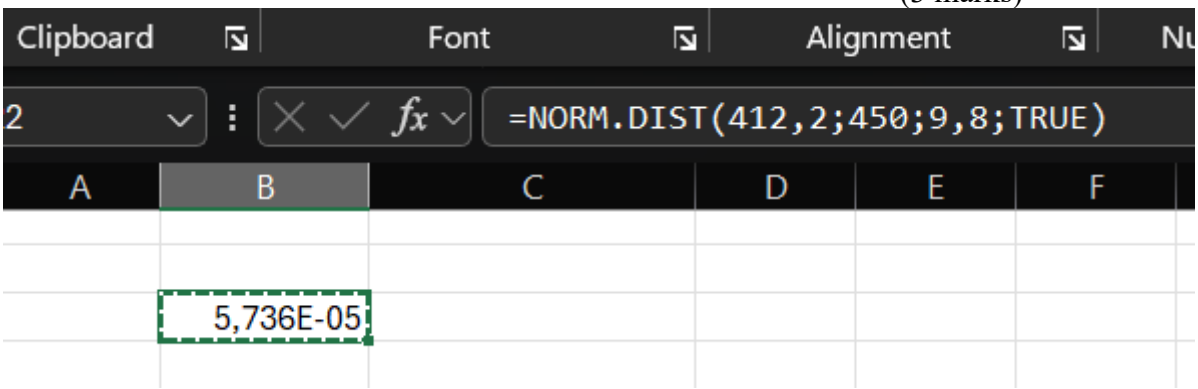
Distribution: Normal
Mean: 450 gr
Standard deviation: 9.8 gr

To calculate standard deviation: knowing the standard deviation for a fillet is 4 gr: $6 * 4^2 = 96$ then $\sqrt{96} = 9.798$. This is way to get standard deviation from a single fillet and summing understanding they are independent.

ii) **Determine the probability that a bulk pack is inadequate.** Use this probability to **explain** if the company is likely to be compliant with Rule 2.

Hints: First work out the weight when the bulk pack becomes inadequate (advertised amount minus tolerance deficiency). Use the distribution from (i) and Excel to calculate the probability.

(3 marks)



The tolerance deficiency is 12.8 gr.The limit is 450-12.8 = 412.2 gr

Appliying Normal distribution function in excel I get that the probability of have an inadequate bulk pack is 0.0000573601 = 0.005736%.

In other words, the probability is 0.01%, the company is compliant with rule 2.

c) Suppose you are now an inspector from the Department of Industry, Science and Resources. The company has agreed to use the 'average method'. Under the 'average method' inspector takes a sample of size 12 (or more if they feel the need).

Describe a method of selecting the sample. Justify decisions you make. State any assumptions you are making.

Hints and marking guide: describe a **suitable sampling plan** (simple random sampling, stratified random sampling, cluster sampling etc) and **why** that would be appropriate. State at least **one source of potential bias** that could arise and **how** the sampling method mitigates the risk. Your response should refer to the context of this situation and not be generic.

(4 marks)

A stratified random sampling method can be used to select the inspection sample. The production would divided in strata based on factors that affect weigh, such as shift or production line. From each stratum, a number of packs would be selected randomly using a random app, until a total sample size of at least 12 packs is gotten.

This way all major production conditions are represented in the sample and minimise the risk of selecting packs only from one shift or one machine, that could change the results.

There is a risk of selecting packs that are easier to access (like the ones are on top of a pallet or from a single batch). Random selection within each stratum helps mitigate this risk.

Assumptions include that production within each stratum is consistent and the random selection process is applied correctly without substitution.

Question 3 (14 marks)

A mobile phone manufacturer tests a random sample of 800 phones and finds 32 are defective.

a. Based upon this sample, name and give the value of the statistic that would be management's best estimate (or a point estimate) for the proportion of the defective phones.

(2 marks)

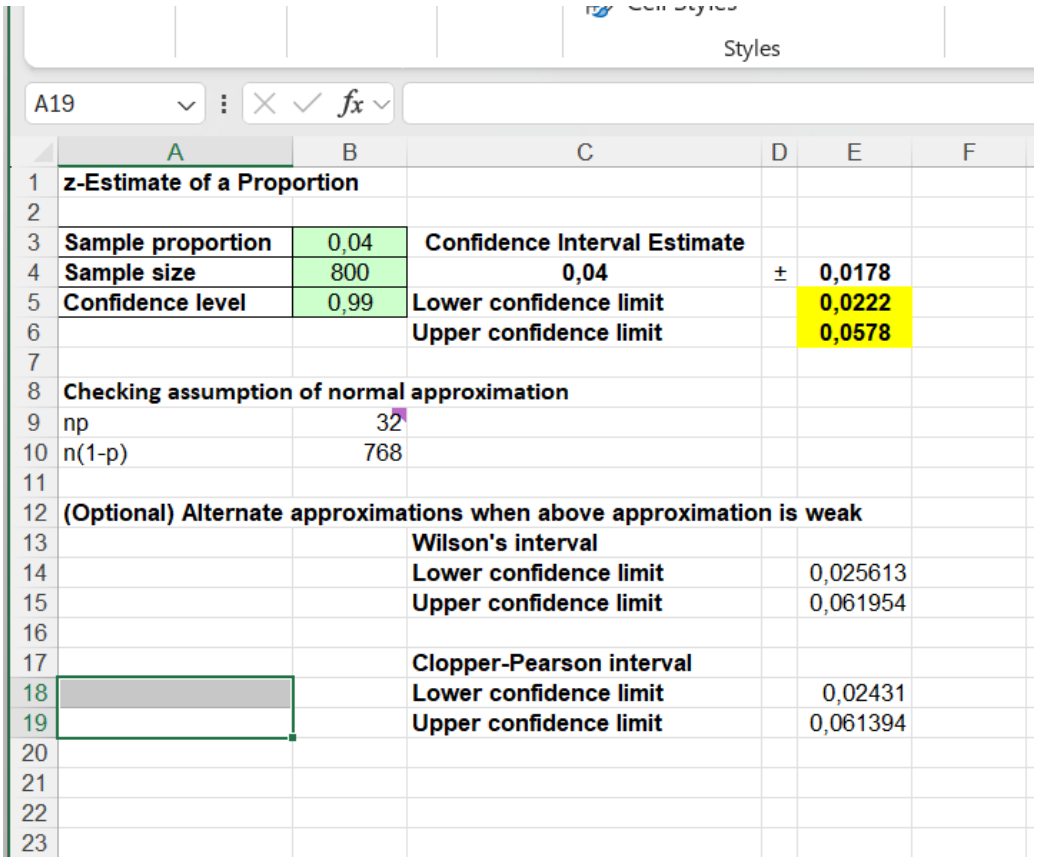
name: sample proportion
value: 4%

b. Use the Excel Estimators Workbook to determine a 99% confidence interval for the proportion of the defective phones. Show that the assumption/s underlying the method used are satisfied. (4 marks)

The sample contains 800 phones, of which 32 are defective. The estimated proportion of defective phones is therefore 4%.

Using the Estimators Workbook, the 99% confidence interval for the proportion of defective phones is between **2.2% and 5.8%**.

The assumptions for using this method are satisfied because the sample size is large and there are enough defective and non-defective phones in the sample. This makes the normal approximation appropriate.



	A	B	C	D	E	F
1	**z-Estimate of a Proportion**					
2						
3	**Sample proportion**	0,04	**Confidence Interval Estimate**			
4	**Sample size**	800	**0,04**	±	**0,0178**	
5	**Confidence level**	0,99	**Lower confidence limit**		**0,0222**	
6			**Upper confidence limit**		**0,0578**	
7						
8	**Checking assumption of normal approximation**					
9	np	32				
10	n(1-p)	768				
11						
12	**(Optional) Alternate approximations when above approximation is weak**					
13			**Wilson's interval**			
14			**Lower confidence limit**		0,025613	
15			**Upper confidence limit**		0,061954	
16						
17			**Clopper-Pearson interval**			
18			**Lower confidence limit**		0,02431	
19			**Upper confidence limit**		0,061394	
20						
21						
22						
23						

c. What is the width of the 99% confidence interval?

(1 mark)

5.8% - 2.2% = 3.6% is the width of the 99% confidence interval

d. Say what would need to be done to get a narrower 99% confidence interval? Explain why that narrows the interval.

(1+1 marks)

To get a narrower 99% confidence interval, the company need to increase the sample size and test more phone. A larger sample reduces random sampling variation, so it becomes more precise and the interval becomes narrower.

Subsequently management finds that the research team in fact selected sample of 800 phones at random from one production shift.

e. Comment briefly (using relevant statistical terminology) on:
   i. the validity of the sampling used and any implications for this inference; and
   ii. whether selecting a larger sample size **in this same manner** would likely lead to more reliable inferences.

(3+2 marks)

i.
If the sample is taken from only one production shift, so it is not representative of the entire factory. Different shifts may have differente machines and operator working in different conditions, what means it can affect the defect rate. It will introduces potential sampling bias, meaning the results may not be as accurate as the true defect rate of all production.
ii. > Taking larger sample from same shift would reduce random error, but it would not remove the bias in any case. The sample represent only one shift and not he whole production process. Therefore, increasing the sample size would not necessarily improve the true about the defect rate.

END OF ASSIGNMENT